1. hwcost round 2: run records (runs started 20261016T1654Z)

Kind: benchmark and analysis runs, descriptive report (observations only; interpretation is in the study report `../../report.pdf`).
Study: `research/hwcost_round2_20261016/`, protocol `PROTOCOL.md` frozen at commit `597502b4d` before parts A–D ran; addendum 1 frozen at `4e5994fc9` before runs cm-1 and cm-2.
Commands: the protocol's "Commands" sections, verbatim. Every run below exited 0 and completed every requested arm (7 repetitions of about 100 ms each, batches 16, 64, 256, 1024, 4096, 16384, 65536, 262144, 1048576, 4194304); no arm failed or timed out.

1.1. Runs

run	binary	fields	started UTC	finished UTC	checkout HEAD (`commit.txt`)
isa-base-1	`target` (default x86-64)	fp64, fp64bf, w1, w2, w3, w4	20261016T1654Z	20261016T1657Z	`597502b4d`
isa-native-1	`target-native` (`-C target-cpu=native`)	fp64, fp64bf, w1, w2, w3, w4	20261016T1657Z	20261016T1700Z	`b427150af`
isa-base-2	`target` (default x86-64)	fp64, fp64bf, w1, w2, w3, w4	20261016T1700Z	20261016T1702Z	`b427150af`
isa-native-2	`target-native` (`-C target-cpu=native`)	fp64, fp64bf, w1, w2, w3, w4	20261016T1702Z	20261016T1705Z	`b427150af`
cm-1	`target` (default x86-64)	fp64, fp64cm	20261016T1722Z	20261016T1723Z	`4e5994fc9`
cm-2	`target` (default x86-64)	fp64, fp64cm	20261016T1723Z	20261016T1724Z	`4e5994fc9`

`commit.txt` is the checkout's HEAD when `measure` started. Both part-A binaries were built at 16:54 UTC from the protocol commit's tree; `b427150af` (committed during the runs) changes only claim printing in `compare`, `gpu-check` and `ic-price` and `fit.rs`'s claim helpers: the timed code (`kernel.rs`, `measure.rs`, `mod.rs`, `rho.rs`) is identical at both commits (`git diff --stat 597502b4d b427150af`). cm-1 and cm-2 ran a baseline binary built at `4e5994fc9`.

Compiled target features: baseline runs none probed (x86-64 baseline) native runs sse4.2, avx, avx2, fma, bmi1, bmi2, adx, avx512f, avx512ifma, pclmulqdq, vpclmulqdq.

1.2. Host and isolation

- CPU: Intel(R) Xeon(R) Processor @ 2.10GHz (GenuineIntel, signature 6/207/2), 4 logical CPUs, 1 NUMA node; virtualization: cpu_flag = hypervisor, systemd_detect_virt_vm = kvm kernel 6.18.44-fc-v139 rustc 1.97.0 (2d8144b78 2026-07-07).
- Caches (sysfs): L1 Data 48 KiB, L2 Unified 2048 KiB, L3 Unified 266240 KiB.
- Isolation **L0**: shared cloud container, no pinning, no frequency control; arms interleaved, order rotated per repetition. Host class `ECBENV2h55b1cb8b1d10`.
- **No hardware performance counters:** the guest kernel reports `Performance Events: unsupported CPU family 6 model 207 no PMU driver, software events only.` Branch misses, retired instructions and cycles could not be counted. This is missing data, not a measured zero.

1.3. Correctness gates (before any timing)

Every run's `stderr.txt` opens with one line per field: the batched walk equals the unbatched reference and stays on the curve, and the timed arithmetic equals the field's (for `fp64bf` and `fp64cm`: the walk is `Fp64`'s point for point, claim F0). All gates passed in all six runs; the unit test `branch_free_fp64_is_bit_for_bit_fp64` compares 10,000 random operations of each variant with `Fp64`.

1.4. Part A: baseline vs native build

Validation claims per run (median |relative error|; threshold 0.30; C0 threshold 0.05). V1 = calibrated model on test batches B ≥ 4096, V2 = a priori model, V3 = size-extrapolated model.

claim	isa-base-1	isa-native-1	isa-base-2	isa-native-2
C0	0.0028 PASS	0.001 PASS	0.007 PASS	0.0028 PASS
V1.fp64	0.0417 PASS	0.0898 PASS	0.0077 PASS	0.0706 PASS
V1.fp64bf	0.0228 PASS	0.0806 PASS	0.0322 PASS	0.0496 PASS
V1.w1	0.0635 PASS	0.0269 PASS	0.0211 PASS	0.0803 PASS
V1.w2	0.0685 PASS	0.1199 PASS	0.0418 PASS	0.075 PASS
V1.w3	0.0146 PASS	0.0219 PASS	0.0262 PASS	0.0454 PASS
V1.w4	0.0625 PASS	0.0774 PASS	0.029 PASS	0.0453 PASS
V2.fp64	0.5486 FAIL	0.1451 PASS	0.5358 FAIL	0.0816 PASS
V2.fp64bf	0.5298 FAIL	0.0953 PASS	0.5505 FAIL	0.0634 PASS
V2.w1	0.0404 PASS	0.0235 PASS	0.0314 PASS	0.043 PASS
V2.w2	0.2032 PASS	0.2313 PASS	0.1777 PASS	0.1511 PASS
V2.w3	0.1713 PASS	0.1453 PASS	0.1378 PASS	0.1039 PASS
V2.w4	0.1356 PASS	0.0843 PASS	0.1306 PASS	0.0671 PASS
V3.w4	0.0938 PASS	0.058 PASS	0.0791 PASS	0.0675 PASS

Calibration transfer (I3): native op costs plus isa-base-1's fitted residuals (`hwcost fit --residual-from`), median |error| on test batches:

claim	isa-native-1	isa-native-2
V1.fp64	1.5061 FAIL	1.3843 FAIL
V1.fp64bf	1.3207 FAIL	1.2942 FAIL
V1.w1	0.0185 PASS	0.0178 PASS
V1.w2	0.0148 PASS	0.0942 PASS
V1.w3	0.0374 PASS	0.0492 PASS
V1.w4	0.0294 PASS	0.0219 PASS

Per-arm comparison (I1) (`runs/compare/compare.csv`, 148 arms): $r_k = B_k/A_k$, drift $d = |A_2/A_1 - 1|$, an effect called only when both ratios clear $\max(2d, 0.02)$ on the same side. Counts per arm group:

group	class	arms
clock	no detected effect	2
fp64	candidate slower	3
fp64	no detected effect	5
fp64bf	candidate slower	1
fp64bf	no detected effect	7
kernel	candidate faster	22
kernel	candidate slower	1
kernel	no detected effect	37
mem	candidate faster	9
mem	candidate slower	1
mem	no detected effect	10
scaling	candidate faster	9
scaling	no detected effect	9
w1	candidate slower	3
w1	no detected effect	5
w2	candidate faster	1
w2	no detected effect	7
w3	candidate faster	1
w3	no detected effect	7
w4	candidate faster	1
w4	candidate slower	3
w4	no detected effect	4

Kernel arms at four batch sizes (ns per rho iteration, medians of 7):

arm	A1	B1	A2	B2	r1	r2	drift	margin	class
fp64.262144	48.3	20.7	48.1	22.2	0.429	0.462	0.004	0.02	candidate faster
fp64.4096	47.9	20.9	47.7	20.1	0.435	0.42	0.004	0.02	candidate faster
fp64.4194304	50.9	22.1	50.6	23.3	0.435	0.46	0.006	0.02	candidate faster
fp64.64	56.5	25.6	55.7	25.6	0.454	0.46	0.015	0.029	candidate faster
fp64bf.262144	49.3	20.3	47.9	20.7	0.412	0.433	0.028	0.057	candidate faster
fp64bf.4096	47.2	20	47.1	20.8	0.423	0.442	0.002	0.02	candidate faster
fp64bf.4194304	52.3	22.5	51.1	22.6	0.431	0.443	0.022	0.045	candidate faster
fp64bf.64	53.7	26	55.4	26.1	0.484	0.47	0.032	0.064	candidate faster
w1.262144	33.9	31.8	31.2	32	0.937	1.028	0.081	0.163	no detected effect
w1.4096	30.1	32.3	30.9	31.4	1.073	1.018	0.024	0.047	no detected effect
w1.4194304	34.3	32.9	33.3	33.3	0.959	1	0.028	0.057	no detected effect
w1.64	40.8	39.7	41	40.5	0.971	0.988	0.004	0.02	no detected effect

w2.262144	79.9	85.5	86.5	80.3	1.071	0.929	0.083	0.167	no detected effect
w2.4096	79.6	84.7	78.9	75.8	1.063	0.961	0.01	0.02	no detected effect
w2.4194304	81.4	87.9	86.4	87.2	1.08	1.009	0.062	0.123	no detected effect
w2.64	108.1	96.3	101.8	96.4	0.891	0.947	0.058	0.116	no detected effect
w3.262144	145.5	144.7	146.3	143.5	0.994	0.981	0.005	0.02	no detected effect
w3.4096	143.9	139	137.1	136.7	0.966	0.997	0.047	0.094	no detected effect
w3.4194304	144.1	151.6	148.5	147.1	1.052	0.99	0.031	0.061	no detected effect
w3.64	179.6	173.5	172.1	173.4	0.966	1.008	0.042	0.083	no detected effect
w4.262144	217.3	228.1	216	208.8	1.05	0.966	0.006	0.02	no detected effect
w4.4096	199.9	211.7	204.5	207.1	1.059	1.012	0.023	0.046	no detected effect
w4.4194304	218.8	225	216	218.1	1.029	1.01	0.013	0.025	no detected effect
w4.64	260.6	250.3	260.8	248.8	0.96	0.954	0.001	0.02	candidate faster

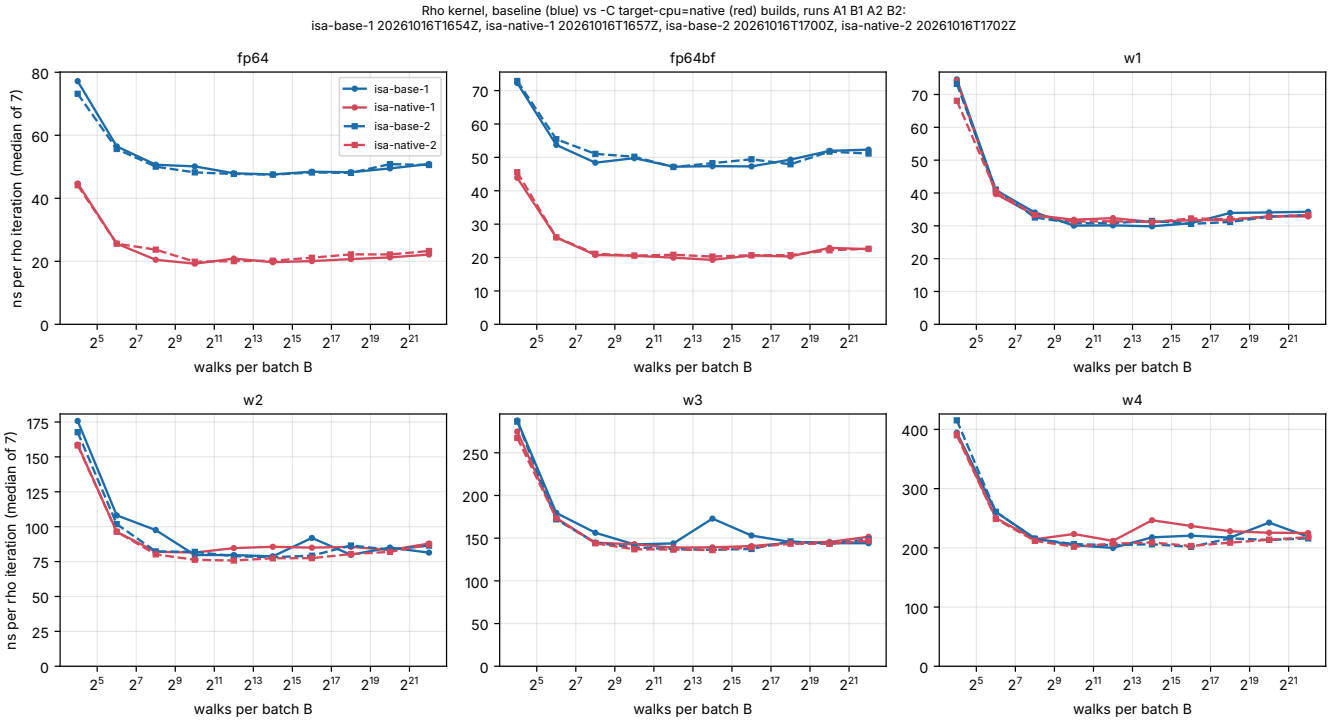


Figure 1: Rho kernel time per field against batch size, runs isa-base-1, isa-native-1, isa-base-2, isa-native-2 (medians of 7 repetitions each).

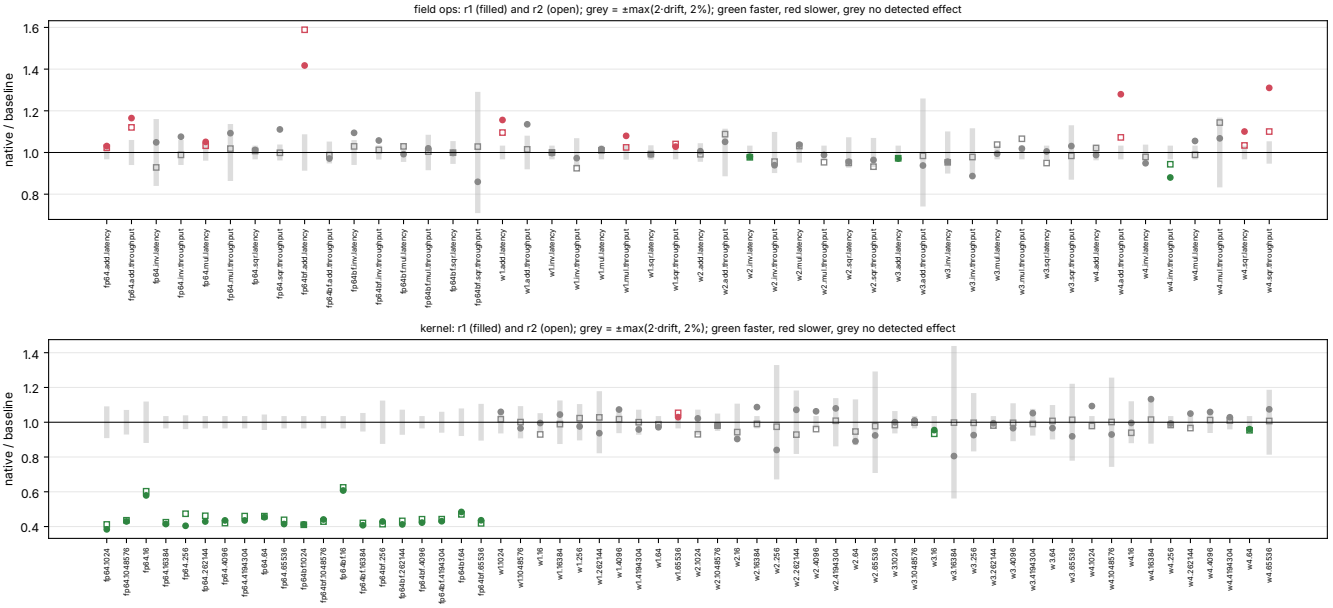


Figure 2: Native / baseline ratio per arm, r_1 and r_2, with the drift margin, all 148 arms of `compare.csv`.

1.5. Part B: `fp64` vs `fp64bf` (baseline runs)

`fp64bf / fp64` kernel ratio at B ≥ 4096 in isa-base-1 and isa-base-2 (F1 cells):

run	B	fp64 ns	fp64bf ns	fp64bf / fp64
isa-base-1	4096	47.93	47.21	0.9849
isa-base-1	16384	47.57	47.36	0.9957
isa-base-1	65536	48.44	47.3	0.9764
isa-base-1	262144	48.26	49.33	1.0221
isa-base-1	1048576	49.48	51.95	1.0499
isa-base-1	4194304	50.88	52.32	1.0283
isa-base-2	4096	47.73	47.11	0.987
isa-base-2	16384	47.51	48.26	1.0158
isa-base-2	65536	48.18	49.43	1.0259
isa-base-2	262144	48.07	47.93	0.9971
isa-base-2	1048576	50.85	51.71	1.017
isa-base-2	4194304	50.56	51.15	1.0116

F1 (worst fp64bf/fp64 kernel ratio at B ≥ 4096 in the baseline runs (12 cells)): 1.0499, threshold 0.85, FAIL.

F3, disassembly (`runs/f3/`, `f3_disasm.sh`; `objdump -d` of each `#[inline(never)]` kernel function; binaries in `runs/f3/binaries.txt`):

build	function	instructions	cond. jumps	cmov	vector-register	mulx
baseline	`step_fp64`	303	32	4	0	0
baseline	`step_fp64bf`	303	32	4	0	0
baseline	`step_fp64cm`	285	20	12	0	0
native	`step_fp64`	337	22	14	38	12
native	`step_fp64bf`	337	22	14	38	12

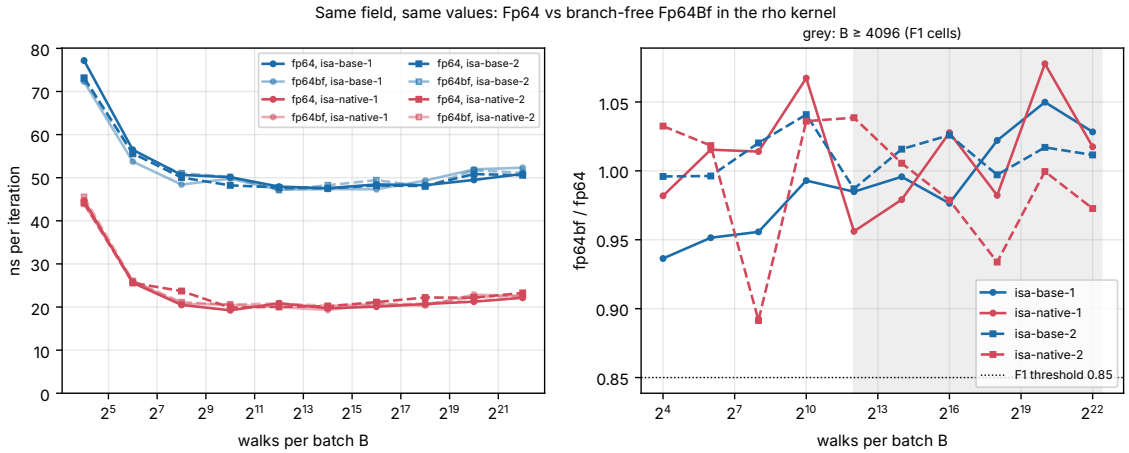


Figure 3: `fp64` and `fp64bf` kernels in all four part-A runs, and their ratio; grey band = F1 cells.

1.6. Addendum 1: `fp64` vs `fp64cm` (runs cm-1, cm-2)

run	B	fp64 ns	fp64cm ns	fp64cm / fp64
cm-1	16	76.79	45.07	0.5869
cm-1	64	55.35	26.07	0.471
cm-1	256	49.71	21.78	0.4383
cm-1	1024	48.18	20.85	0.4327
cm-1	4096	48.41	20.28	0.419
cm-1	16384	46.66	20.36	0.4364
cm-1	65536	48.08	21.27	0.4424
cm-1	262144	48.97	21.25	0.434
cm-1	1048576	49.76	22.16	0.4452

cm-1	4194304	49.21	24.59	0.4998
cm-2	16	72.2	44.17	0.6117
cm-2	64	60.03	26.44	0.4404
cm-2	256	49.27	21.78	0.4421
cm-2	1024	48.33	21.66	0.4482
cm-2	4096	50.96	21.71	0.4261
cm-2	16384	51.27	21.2	0.4135
cm-2	65536	52.7	21.34	0.4048
cm-2	262144	48.48	20.25	0.4176
cm-2	1048576	49.56	21.3	0.4298
cm-2	4194304	50.07	23.08	0.4609

F1prime (worst fp64cm/fp64 kernel ratio at B ≥ 4096 over 2 runs (12 cells)): 0.4998, threshold 0.85, PASS.

claim	cm-1	cm-2
C0	0.009 PASS	0.0385 PASS
V1.fp64	0.0149 PASS	0.031 PASS
V1.fp64cm	0.0587 PASS	0.0679 PASS
V2.fp64	0.5407 FAIL	0.554 FAIL
V2.fp64cm	0.0791 PASS	0.0425 PASS

Fitted residual per iteration (model C): cm-1 fp64 27.2 ns, fp64cm −2 ns; cm-2 fp64 27 ns, fp64cm −2 ns.

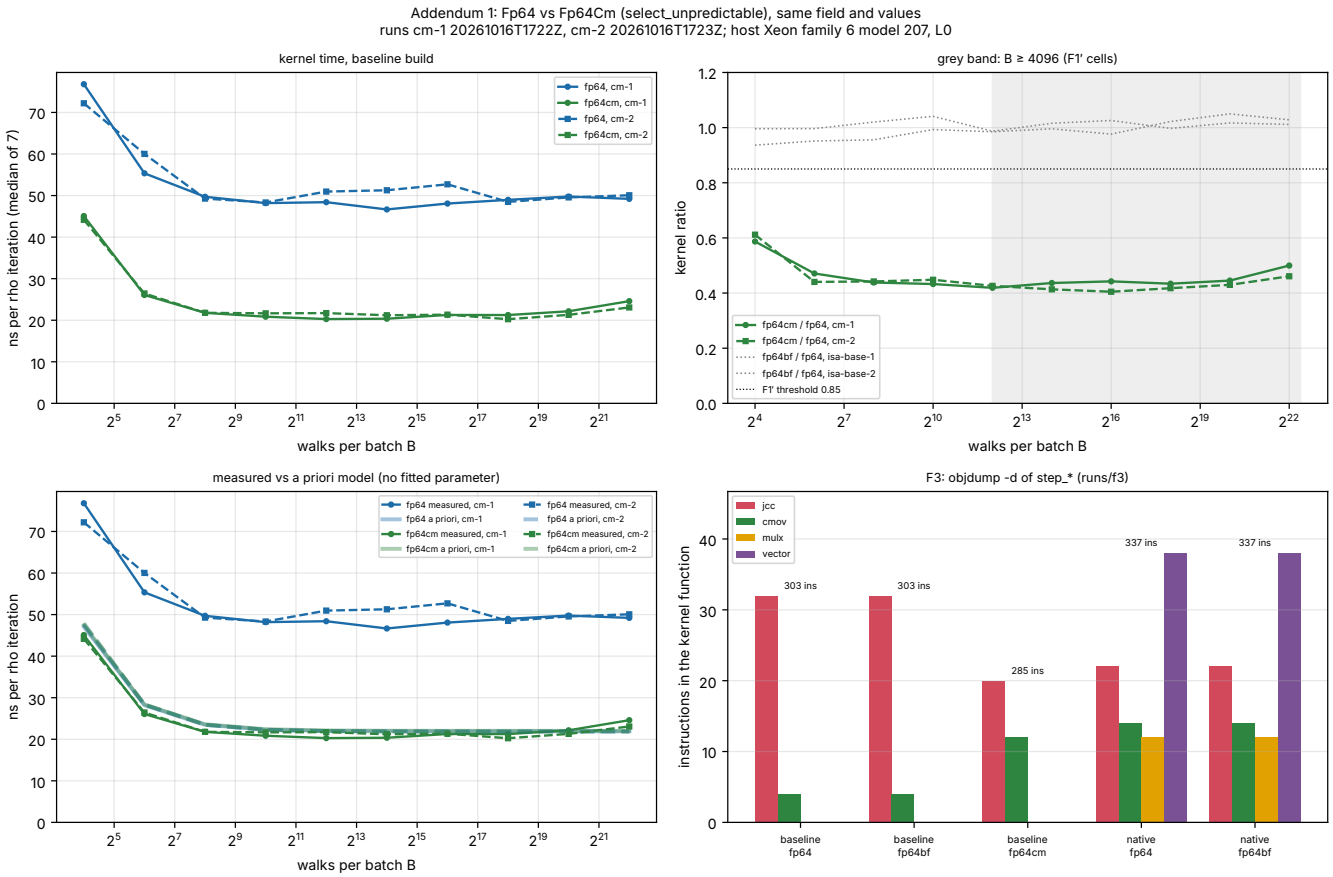


Figure 4: Runs cm-1 and cm-2: kernel time, `fp64cm / fp64` ratio (part A's `fp64bf / fp64` dotted for reference), measured against the a priori model, and the F3 instruction counts.

1.7. Part C: GPU profile against 46 ECC2K-130 builds

Input: `ecc2k130/benchmarks/roofline/calibration.json` (recorded rates; no GPU ran in this study). Output `runs/gpu/` (per-build rows in `gpu_check.csv`).

id	statement	value	threshold	verdict

G1	a priori, all 46 builds, median |error|	0.1792	0.3	PASS
G1.eta>1	builds measured above the a priori ceiling (η > 1)	0	—	REPORT
G2	η from each GPU's base build, 44 other builds, median |error|	0.0145	0.3	PASS
G2.10	the same, against 10%	0.0145	0.1	PASS
G3	η from the RTX PRO 6000 base build carried to 25 B200 builds, median |error|	0.195	0.3	PASS
G1.shared	shared-int pipes: a priori, median |error|	0.1237	—	REPORT
G1.shared.eta>1	shared-int pipes: builds with η > 1	0	—	REPORT
G2.shared	shared-int pipes: η per GPU, median |error|	0.0192	—	REPORT
G3.shared	shared-int pipes: η carried across GPUs, median |error|	0.1072	—	REPORT

Cost-arithmetic check (`summary.txt`): `expected ECC2K-130 work 2^60.9 at 14.63753 B/s on rtx-pro-6000 ($3.031/h): 40827.8 h, $123749`
`expected ECC2K-130 work 2^60.9 at 15.115792 B/s on rtx-pro-6000 ($3.031/h): 39536.1 h, $119834`

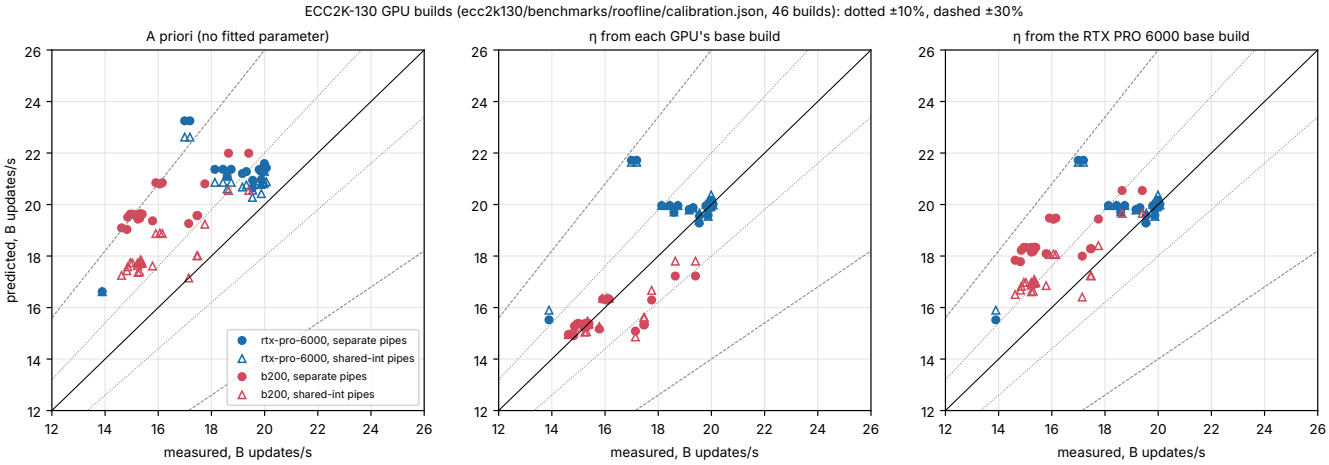


Figure 5: Predicted against measured update rate, 46 builds, both pipe models; the three panels are G1, G2, G3.

1.8. Part D: IC and rho in wall time (frozen campaign records)

Input: `research/ic_frontier_campaign_20261015` Phase 1 sessions and §5a band files (32 counted, verified runs per arm and curve; host Xeon 2.30 GHz, 4 vCPU, L0). No solve ran in this study. Dollars at $0.1785/h ÷ 4 per vCPU-hour (illustrative). The `rank` column applies the §5a rule (wall ratio and all three band ratios on one side of 1) per measured curve; it was added to `ic-price` after the analysis and changes no other value.

koblitz: ic-compact vs rho-frob.

curve	log2 r	IC ms	rho ms	wall IC/rho	band w=0	w=0.1	w=1	in band	w*	rank
K13	11	0.15	3.02	0.049	6.87	9.33	27.1	false	−0.254	unranked
K17	16	0.25	3.07	0.081	3.8	4.91	11.85	false	−0.28	unranked
K19	18	0.32	3.25	0.099	4.88	7.46	23.6	false	−0.165	unranked
K23	22	1.06	3.57	0.296	5.48	7.39	18.45	false	−0.226	unranked
K37	27.8	4.35	4.15	1.049	5.2	8.31	23.29	false	−0.111	IC > rho
K43	32.1	22.77	3.03	7.504	7.35	10.17	23.14	true	0.005	IC > rho
K47	36.6	141.39	13.44	10.52	9.92	12.7	25.28	true	0.02	IC > rho
K41	39	498.9	25.69	19.424	15.02	17.79	30.3	true	0.169	IC > rho

D1.in_band: 3 (curves (of 8) whose measured IC/rho wall ratio lies inside the band's ratio range)
D1.w_star_median: −0.1379 (median implied weight w* (GAE per unpriced native operation))
D1.rank.IC>rho: 4 (curves (of 8) ranked IC > rho (§5a: wall ratio and all three band ratios agree))
D1.rank.IC<rho: 0 (curves (of 8) ranked IC < rho (§5a: wall ratio and all three band ratios agree))
D1.rank.unranked: 4 (curves (of 8) left unranked (wall ratio and band ratios disagree on the order))

Fitted exponents $\log_2 y = a + \alpha \log_2 r$ over the tier: ic wall_ms α = 0.602 (4 pts); ic P(w=0) α = 0.586 (4 pts); ic P(w=0.1) α = 0.551 (4 pts); ic P(w=1) α = 0.497 (4 pts); rho wall_ms α = 0.255 (4 pts); rho P(w=0) α = 0.457 (4 pts); rho P(w=0.1) α = 0.459 (4 pts); rho P(w=1) α = 0.466 (4 pts).

Pre-registered projection (wall-time exponents; extrapolated, not measured):

log2 r	IC core-yr	rho core-yr	wall ratio	band w=0	w=1	rank
48	5.880844e-7	3.129375e-9	187.9	30.5	34.3	IC > rho
64	4.654268e-4	5.311050e-8	8763.4	126.5	48.3	IC > rho
80	3.683521e-1	9.013703e-7	408657.9	524.7	68	IC > rho
96	2.915244e2	1.529770e-5	19056751.1	2176.5	95.7	IC > rho
112	2.307208e5	2.596265e-4	888664447.4	9028.9	134.8	IC > rho
128	1.825991e8	4.406277e-3	41440668188.7	37455.9	189.8	IC > rho

Post hoc projection (`projections_gae.csv`, addendum 1): fitted $P(w)$ exponents at the matched rho's ns per GAE at its largest measured size, log2 r = 128:

w	ns/GAE	IC core-yr	rho core-yr	IC $	IC/rho
0	256.460	5.282051e7	1.410205e3	2.066247e10	37455.9
0.1	233.234	7.167888e6	1.547560e3	2.803954e9	4631.7
1	128.499	4.526957e5	2.385653e3	1.770867e8	189.8

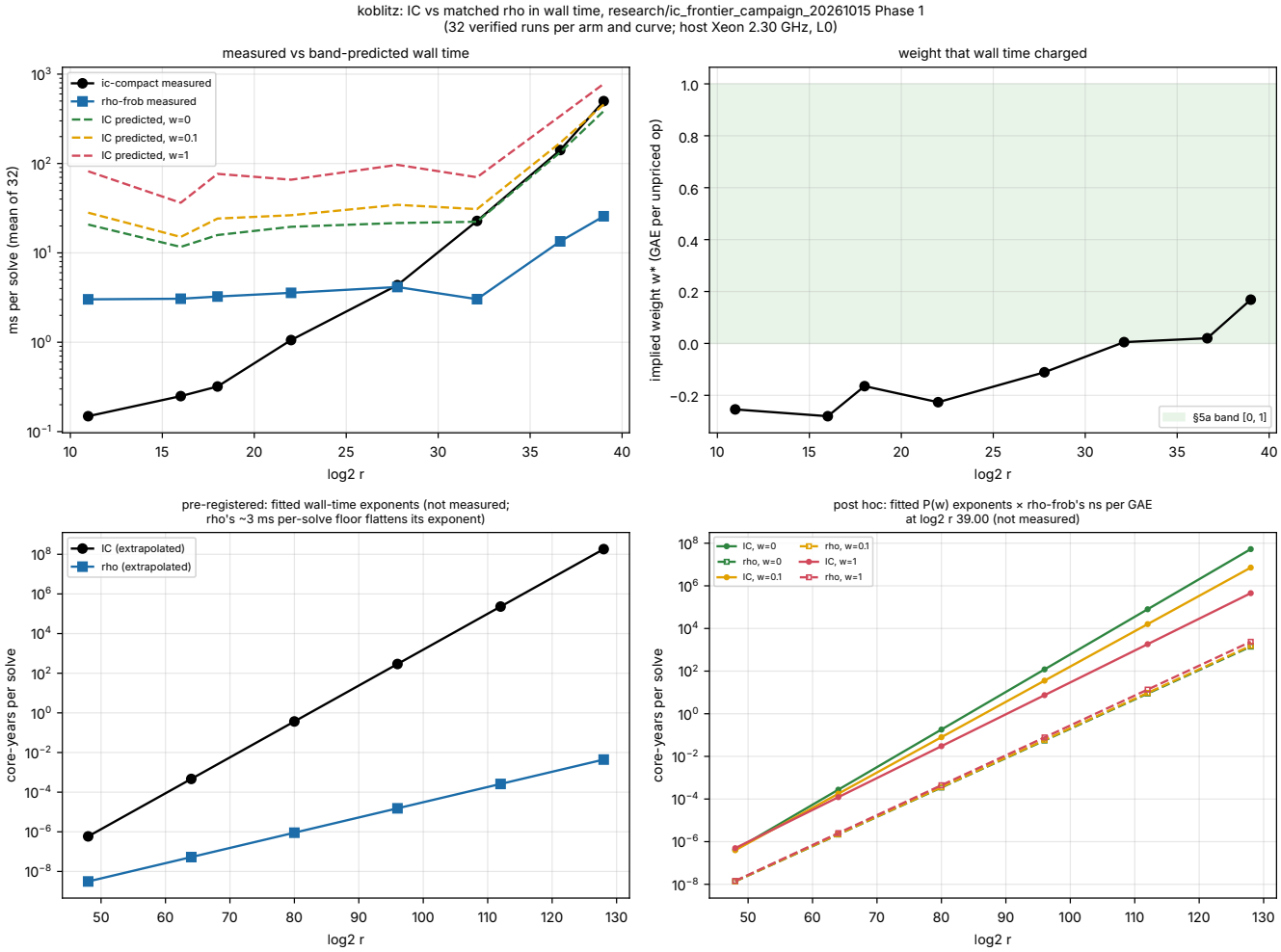


Figure 6: koblitz: measured and band-predicted wall time per solve (mean of 32), implied weight w^*, and the two projections (extrapolations, not measurements).

prime: ic-pair vs rho-neg.

curve	log2 r	IC ms	rho ms	wall IC/rho	band w=0	w=0.1	w=1	in band	w*	rank
P20	20	2.44	3.71	0.658	9.82	10.46	13.51	false	−0.615	unranked
P24	24	8.89	4.44	2.003	12.09	12.23	12.89	false	−0.929	IC > rho
P28	28	59.49	6.66	8.93	18.59	18.64	18.86	false	−0.966	IC > rho
P32	32	544.24	11.68	46.604	40.87	40.91	41.08	false	−1.085	IC > rho
P36	36	4913.35	51.48	95.435	58.15	58.03	57.48	false	−0.967	IC > rho
P40	40	63517.38	171.71	369.907	95.22	101.8	131.41	false	−1.359	IC > rho

D1.in_band: 0 (curves (of 6) whose measured IC/rho wall ratio lies inside the band's ratio range)
D1.w_star_median: −0.9668 (median implied weight w* (GAE per unpriced native operation))
D1.rank.IC>rho: 5 (curves (of 6) ranked IC > rho (§5a: wall ratio and all three band ratios agree))
D1.rank.IC<rho: 0 (curves (of 6) ranked IC < rho (§5a: wall ratio and all three band ratios agree))
D1.rank.unranked: 1 (curves (of 6) left unranked (wall ratio and band ratios disagree on the order))

Fitted exponents $\log_2 y = a + \alpha \log_2 r$ over the tier: ic wall_ms α = 0.799 (5 pts); ic P(w=0) α = 0.659 (5 pts); ic P(w=0.1) α = 0.663 (5 pts); ic P(w=1) α = 0.68 (5 pts); rho wall_ms α = 0.337 (5 pts); rho P(w=0) α = 0.469 (5 pts); rho P(w=0.1) α = 0.47 (5 pts); rho P(w=1) α = 0.472 (5 pts).

Pre-registered projection (wall-time exponents; extrapolated, not measured):

log2 r	IC core-yr	rho core-yr	wall ratio	band w=0	w=1	rank
48	1.385444e-4	2.649696e-8	5228.7	286	376.5	IC > rho
64	9.808659e-1	1.118364e-6	877054.2	2352.4	3766.4	IC > rho
80	6.944329e3	4.720309e-5	147115995.2	19348.1	37680.8	IC > rho
96	4.916442e7	1.992313e-3	24677057813.7	159132.5	376977.5	IC > rho
112	3.480740e11	8.409007e-2	4139299614988.4	1308817.6	3771466.7	IC > rho
128	2.464293e15	3.549212e0	694321074740973.3	10764634.7	37731591.7	IC > rho

Post hoc projection (`projections_gae.csv`, addendum 1): fitted $P(w)$ exponents at the matched rho's ns per GAE at its largest measured size, log2 r = 128:

w	ns/GAE	IC core-yr	rho core-yr	IC $	IC/rho
0	186.197	1.415294e11	1.314763e4	5.536387e13	10764634.7
0.1	169.277	1.920460e11	1.362491e4	7.512507e13	14095215.6
1	93.120	6.053500e11	1.604358e4	2.368025e14	37731591.7

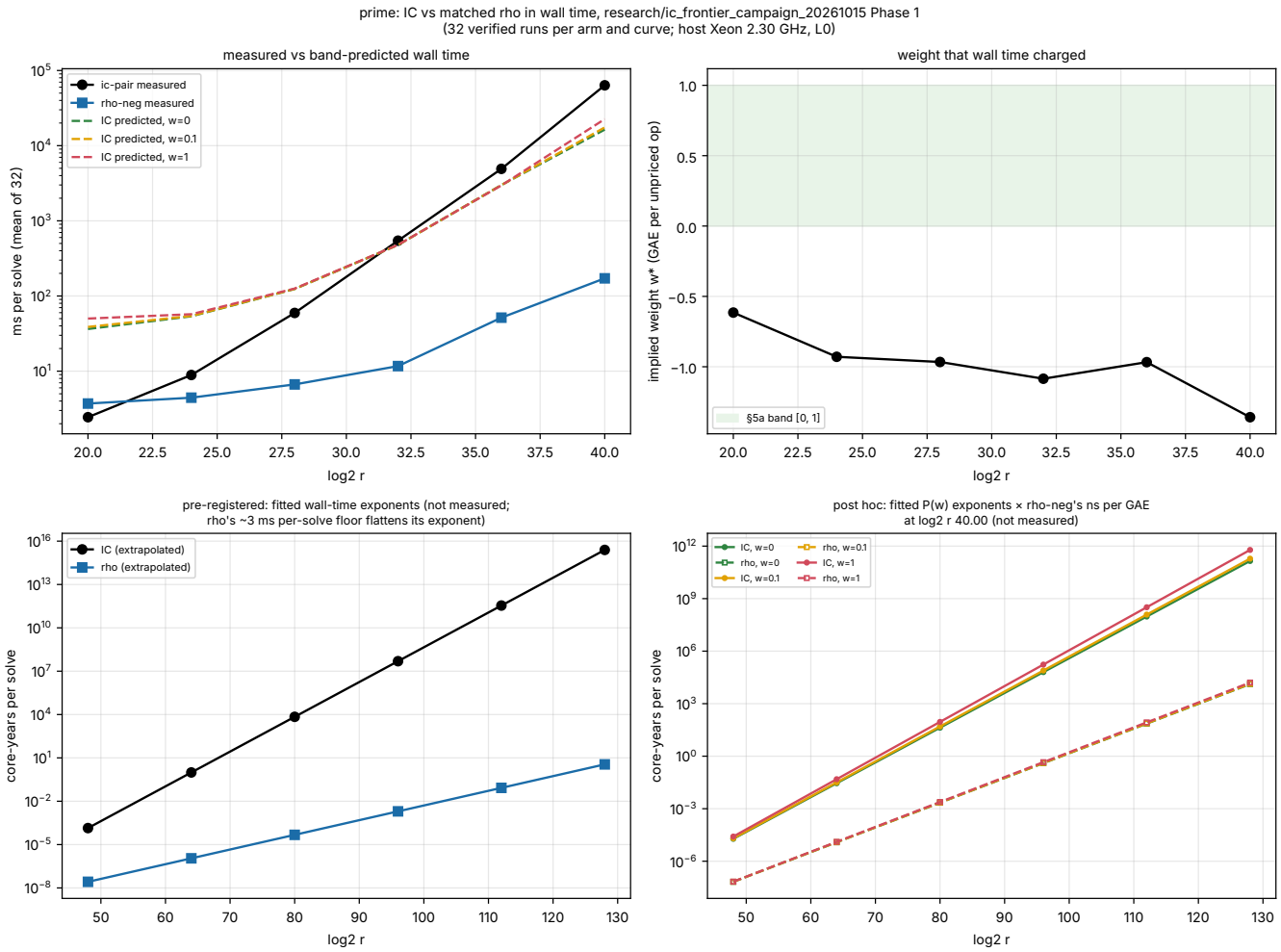


Figure 7: prime: measured and band-predicted wall time per solve (mean of 32), implied weight w^*, and the two projections (extrapolations, not measurements).

1.9. Deviations and missing data

- No PMU: branch-miss counts are missing; part B and addendum 1 test the branch hypothesis by substitution instead.
- F3 found `step_fp64` and `step_fp64bf` identical in the baseline binary, so F1′s two arms ran the same instructions; addendum 1 (frozen before cm-1 and cm-2) added `fp64cm`.
- Part D's rho wall time carries a per-solve floor of about 3–4 ms on the smaller curves (the `rho ms` column); the post hoc GAE projection was added for that reason and is labelled post hoc.
- `run,batch,…` headers of `cm-pair/pairs.csv` name the compared fields (`fp64cm_ns`); the values were unchanged by that fix.

1.10. Raw records

`runs/isa-*/` and `runs/cm-*/`: `measurements.json`, `samples.csv`, `medians.csv`, `validation.{json,csv}`, `profile.json`, `fit.txt`, `stderr.txt`, timestamps, `commit.txt`; `runs/isa-native-*/transfer/`; `runs/compare/`; `runs/cm-pair/`; `runs/f3/`; `runs/gpu/`; `runs/ic-koblitz/`, `runs/ic-prime/`. Graph sources: `../../plot.py` (layout only).